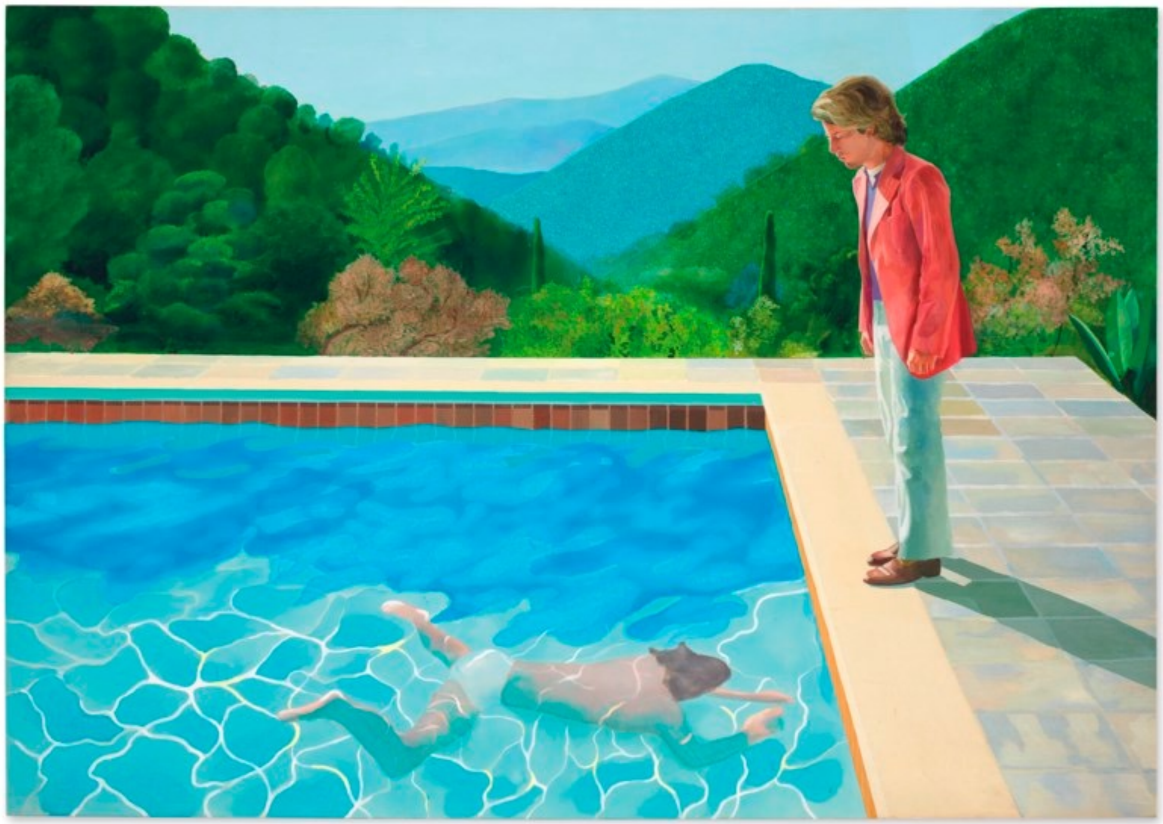
CS 444: Deep Learning for Computer Vision



D. Hockney, *Pool with two figures*, 1972

https://slazebni.cs.illinois.edu/spring25/

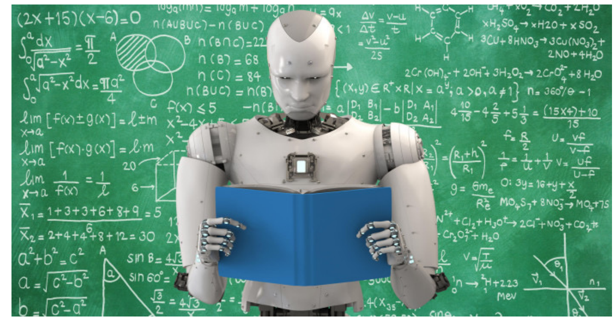
Overview

- Logistics
- The statistical learning framework
- A taxonomy of learning problems
- Topics to be covered in class

How can we build an agent to…

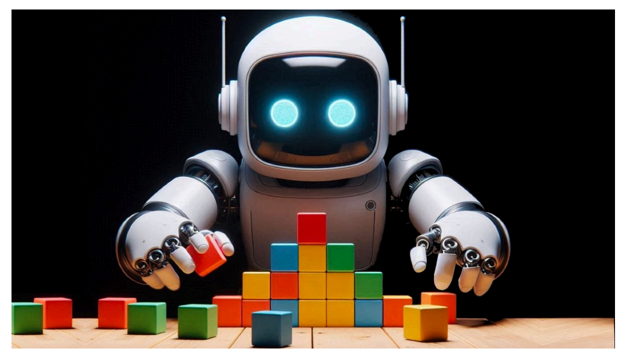
Play games, plan, and reason?

Understand language?



Understand images?

Manipulate the physical world?



How can we build an agent to reach human-level expertise?

- GOFAI (Good old-fashioned AI) answer: figure out how human experts solve the task and program the expertise into the agent

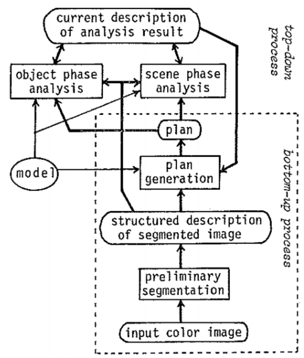


Figure 1. Flow of data in the analysis process.

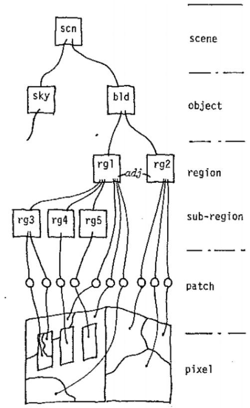


Figure 3. Structure of description built by the system.

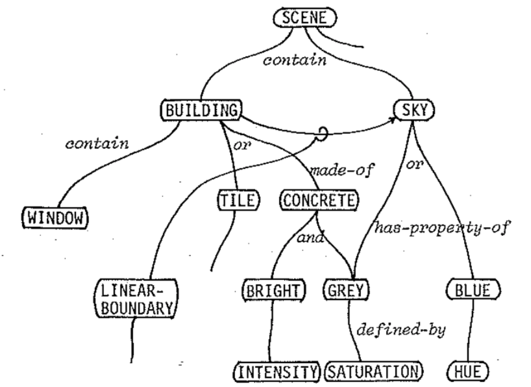


Figure 2. Semantic network for knowledge representation.

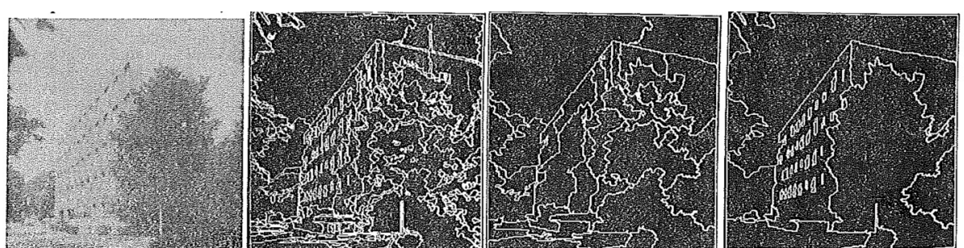


Figure 5-a. Digitized color scene. 5-b. Result of preliminary segmentation. 5-c. Plan image. 5-d. Result of semantic segmentation.

Y. Ohta, T. Kanade and T. Sakai. [An Analysis System for Scenes Containing objects with Substructures.] Proc. of the Fourth International Joint Conference on Pattern Recognition, pp. 752-754, 1978

How can we build an agent to reach human-level expertise?

- GOFAI (Good old-fashioned AI) answer: figure out how human experts solve the task and program the expertise into the agent

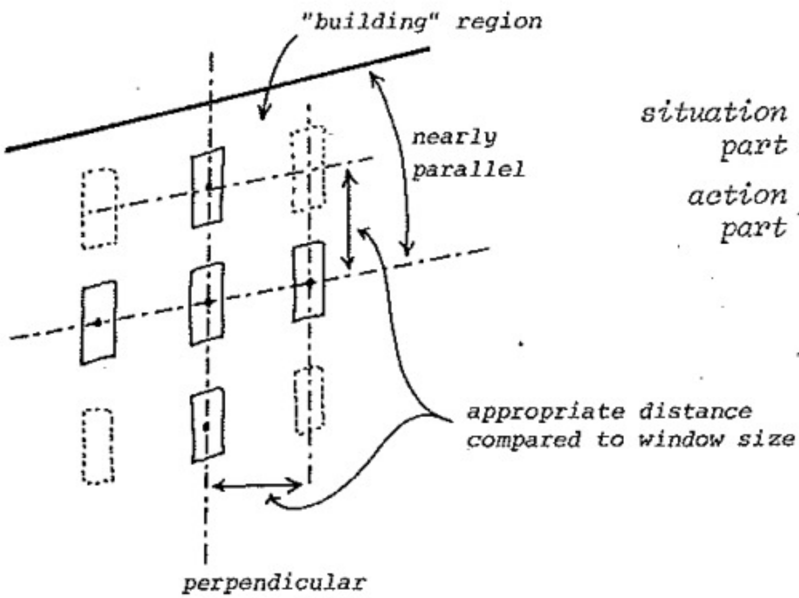


Figure 4-a. "Building" region and "windows".

situation part

action part

```
(ACT              "indicates following is production"
(IF (*WINDOW-LIKE *RGN)
   (THEN (GET-SET *PLSET (PLAN *MRGN) PATCHES)
              "assign to *PLSET all patches which belong to the plan of *MRGN"
      (AND (ALL-FETCH *WLIKE *PLSET
              (AND (IS (LABEL *WLIKE) NIL)(*WINDOW-LIKE *WLIKE)))
           (ALL-FETCH *WIND *WLIKE
              (THERE-IS *WK *WLIKE (*W-RELATION *WIND *WK)))
               "search all patches that have at least one partner patch
                with which *W-RELATION holds"
           (THERE-IS *WK *WLIKE (IS *RGN *WK))))))
(THEN (CONCLUDE P-LABEL B-WINDOW)   "*RGN is 'window'"
   (FOR-EACH *WIND (MUST-BE *WIND P-LABEL B-WINDOW))
              "all patches in *WIND must be 'window' if *RGN is 'window'"
   (DONE-FOR *WIND)   "no need to examine the *WIND patches any more
                       under this context --- for controler"
   (PRIORITY (ADD 2100 (NUMBER *WIND))) ))
              "premium is the number of patches in *WIND"
```

Figure 4-b. The production for analyzing "windows".

Having those phases in the control structure, the production rules can be divided into many groups and the number of rules needed to be examined in each phase is reduced. The rules which perform the overall analysis in the scene phase are described in the knowledge block

How can we build an agent to reach human-level expertise?

- GOFAI (Good old-fashioned AI) answer: figure out how human experts solve the task and program the expertise into the agent
  - Never worked (in general)

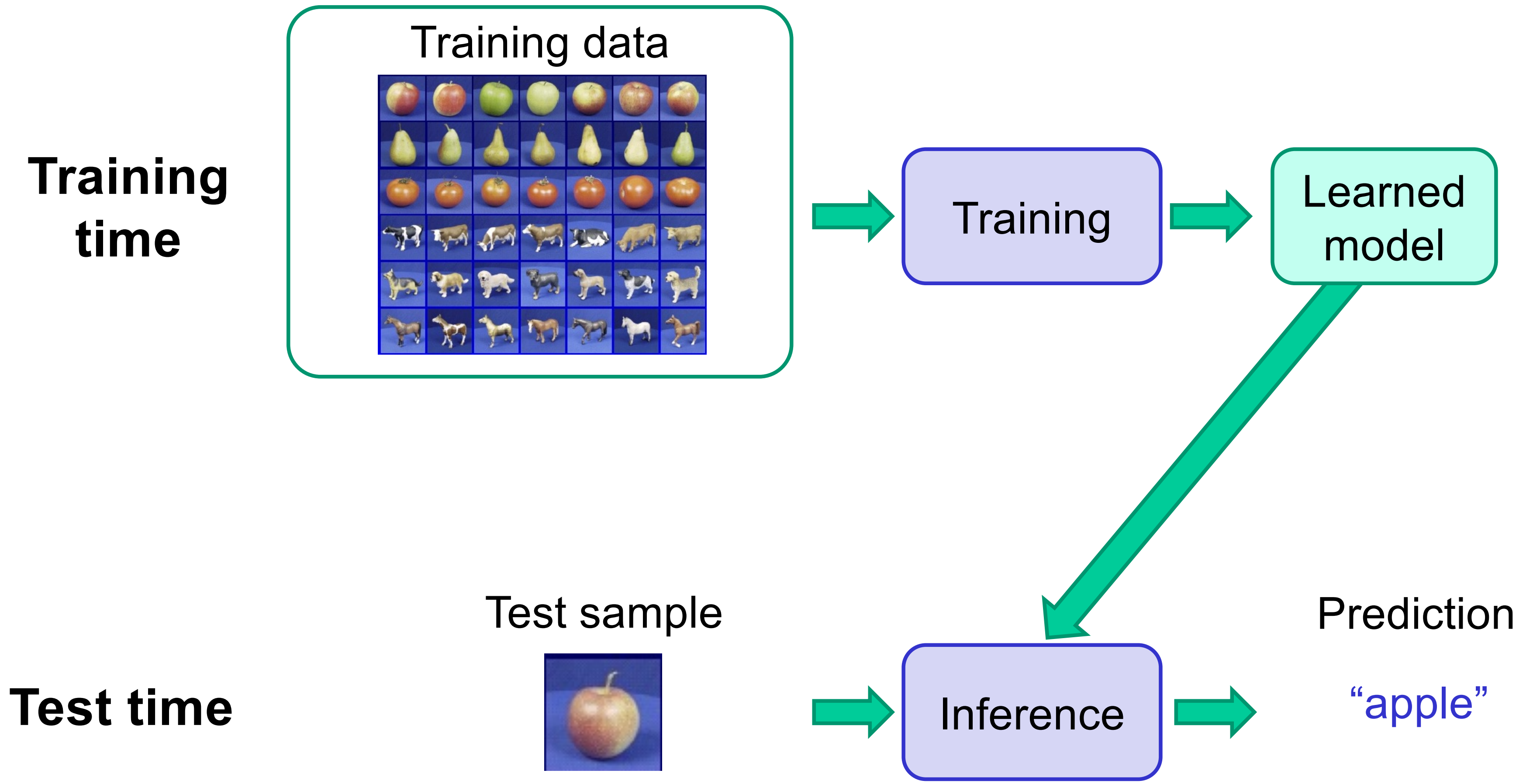
Appendix-B Complete Listing of the Model

How can we build an agent to reach human-level expertise?

- GOFAI (Good old-fashioned AI) answer: figure out how human experts solve the task and program the expertise into the agent

- Modern answer: Program into the agent the **ability to improve performance based on experience**
  - **Ability to improve:** learning algorithm
  - **Performance:** quantified using some score or metric (loss, reward, etc.)
  - **Experience:** training data or demonstrations

The statistical learning framework



The statistical learning framework (supervised)

$$y = f(x)$$

output prediction function input

- **Training** (or **learning**): given a training set of *labeled* examples $\{(x_1, y_1), \ldots, (x_N, y_N)\}$, instantiate a predictor f
- **Testing** (or **inference**): apply f to a new test example x and output the predicted value $y = f(x)$

Overview

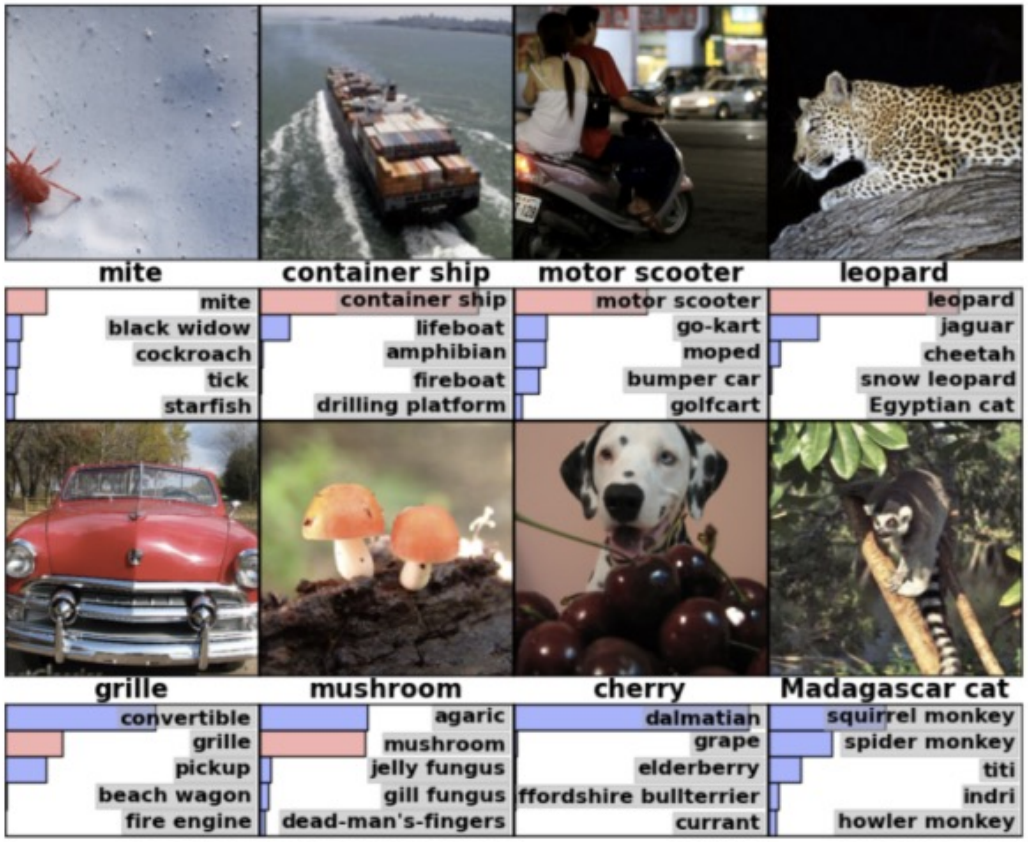
- Logistics
- The statistical learning framework
- A taxonomy of learning problems

Taxonomy of learning problems

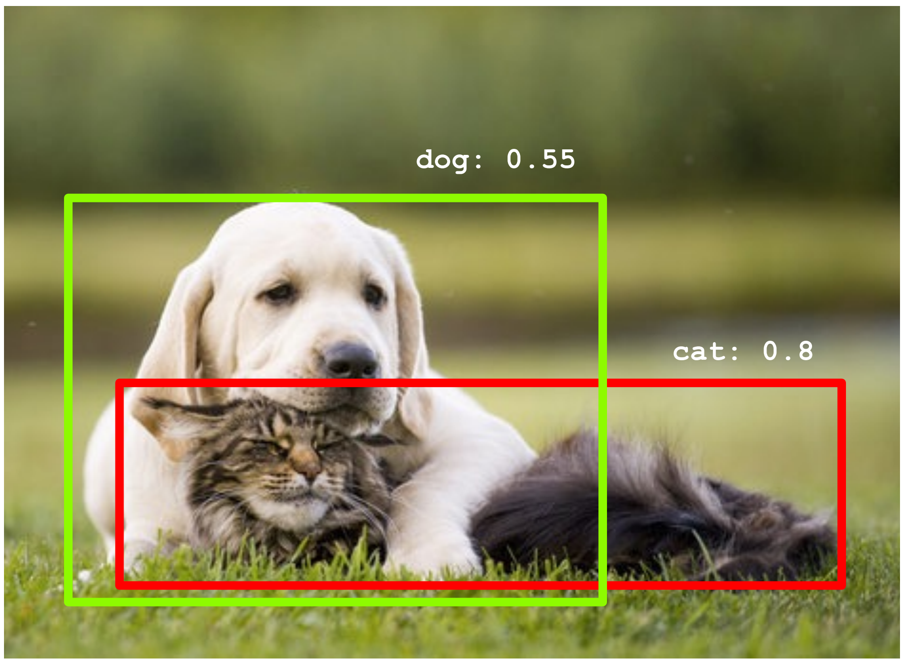
- **Type of prediction problem**
  - Classification
  - Regression
  - Structured prediction
  - Dense prediction
  - Multi-modal prediction
- **Type of supervision**
  - Fully supervised
  - Unsupervised
  - Self-supervised or predictive learning
- **Training regime**
  - Batch offline learning
  - Online/continual learning
  - Active learning
  - Reinforcement learning

Type of prediction problem: Classification

[ImageNet Large-Scale Visual Recognition Challenge (ILSVRC)]



Type of prediction problem: Object detection



- For each bounding box:
    - Coordinates, e.g. (x_0, y_0, w, h)
    - Object category
    - Confidence

Type of prediction problem: Dense prediction

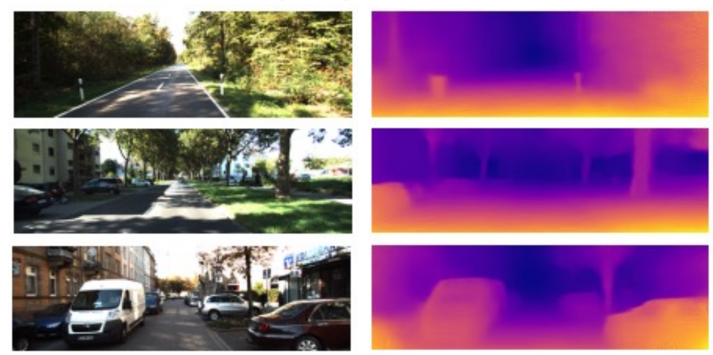
Semantic segmentation

Long et al. (2016)

Image colorization

Zhang et al. (2016)

Depth prediction



Wang et al. (2017)

Type of prediction problem: Dense + structured prediction

Object detection + instance segmentation

Keypoint detection

K. He, G. Gkioxari, P. Dollar, and R. Girshick, [Mask R-CNN](), ICCV 2017

Type of prediction problem: Image description

"man in black shirt is playing guitar."

"construction worker in orange safety vest is working on road."

"two young girls are playing with lego toy."

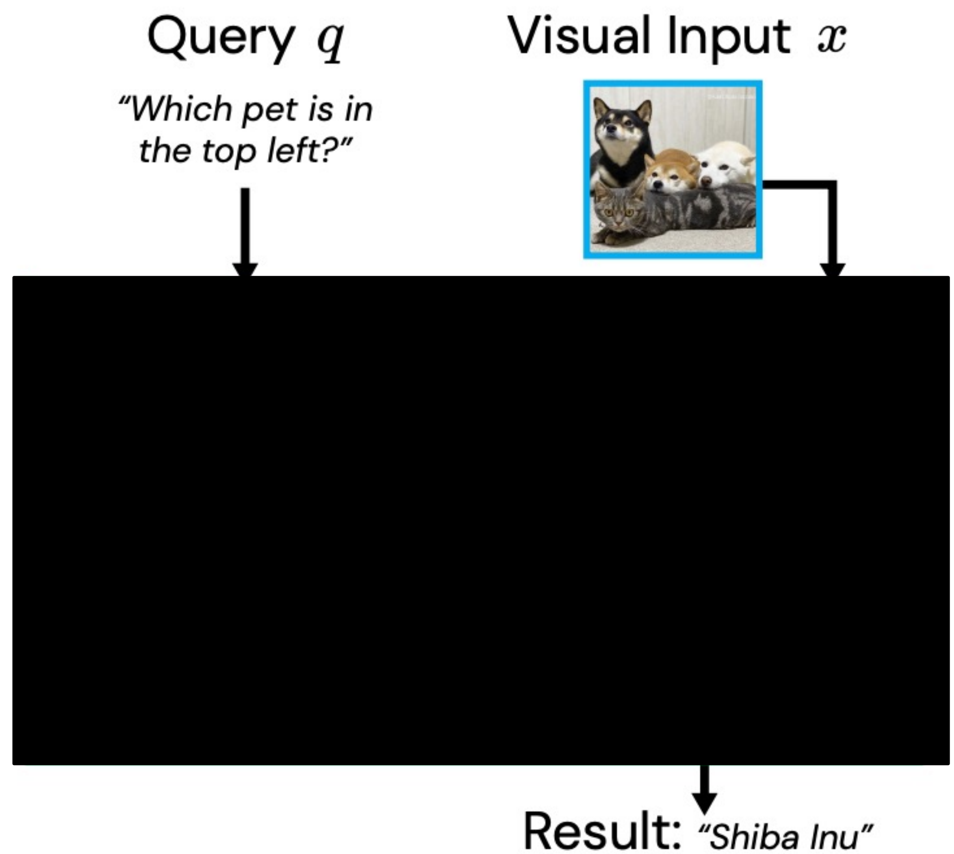
"girl in pink dress is jumping in air."

"black and white dog jumps over bar."

"young girl in pink shirt is swinging on swing."

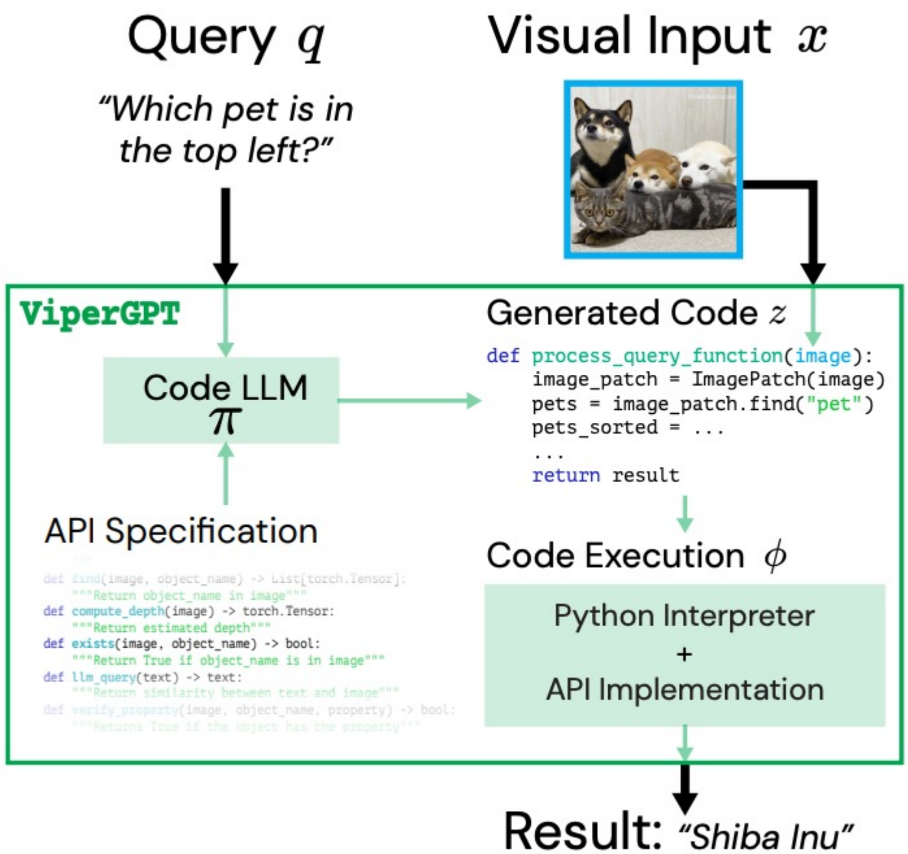
Figure source

Multi-modal prediction



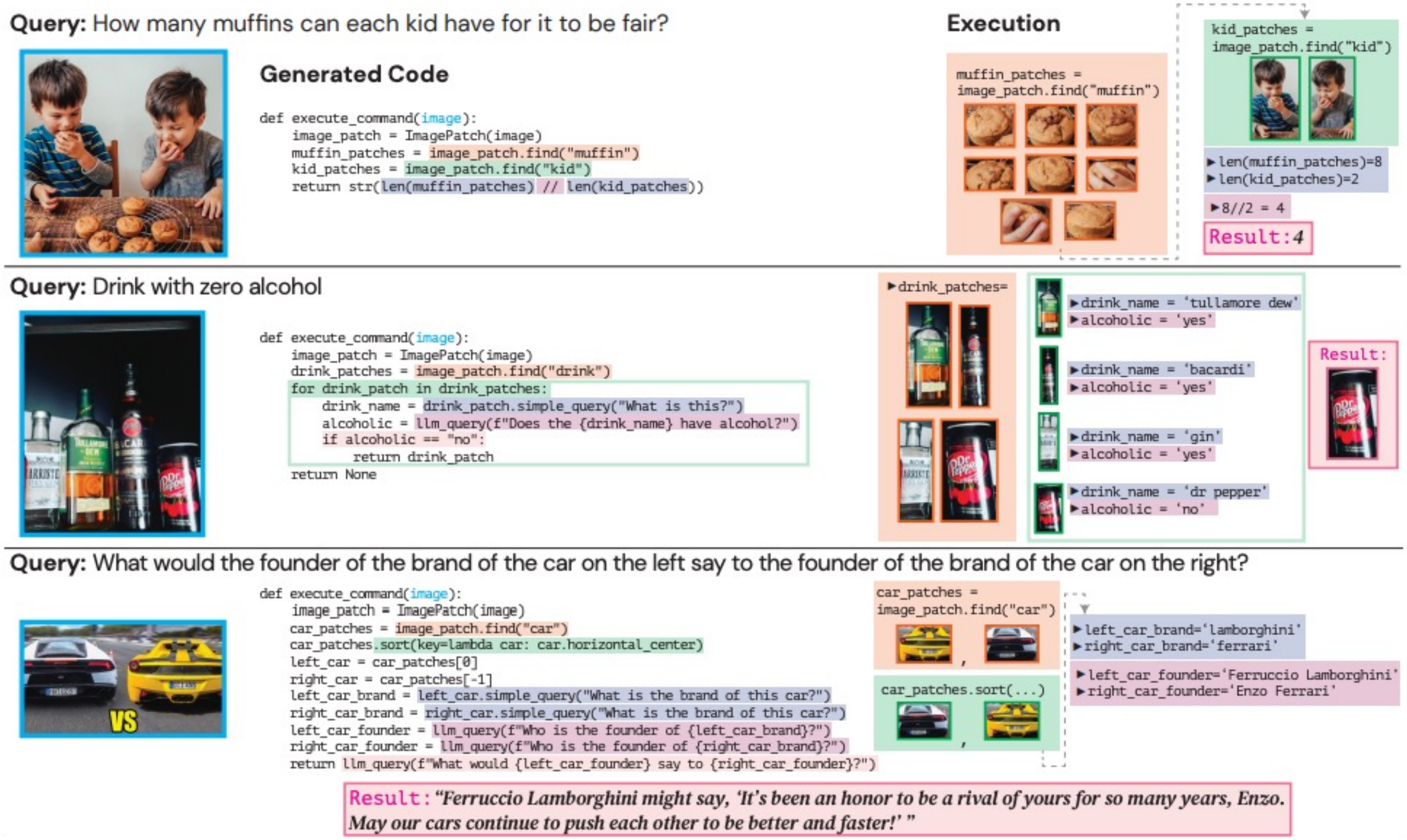
S. Menon et al. [ViperGPT: Visual Inference via Python Execution for Reasoning.](ViperGPT: Visual Inference via Python Execution for Reasoning) ICCV 2023

Multi-modal prediction



S. Menon et al. [ViperGPT: Visual Inference via Python Execution for Reasoning.] ICCV 2023

Multi-modal prediction



S. Menon et al. [ViperGPT: Visual Inference via Python Execution for Reasoning.]() ICCV 2023

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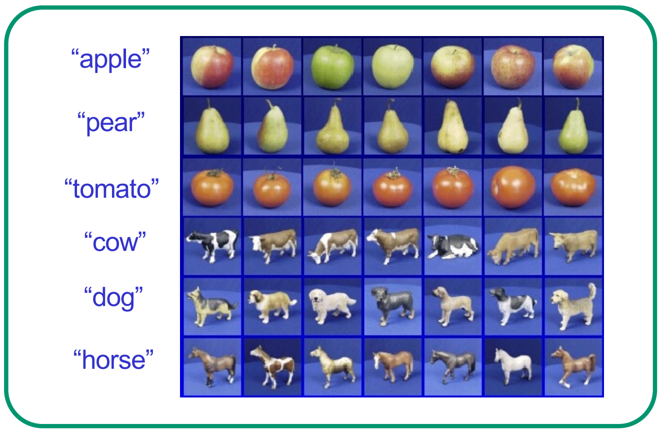
Type of supervision

- Traditional dichotomy: **supervised** vs. **unsupervised** learning

Supervised learning:
clean, complete training labels for the task of interest



Unsupervised learning:
no labels

Unsupervised learning

- Given: large collection of unlabeled data
- Goal: ???

Image source

Unsupervised learning

- Traditional: Clustering
  - Discover groups of “similar” data points

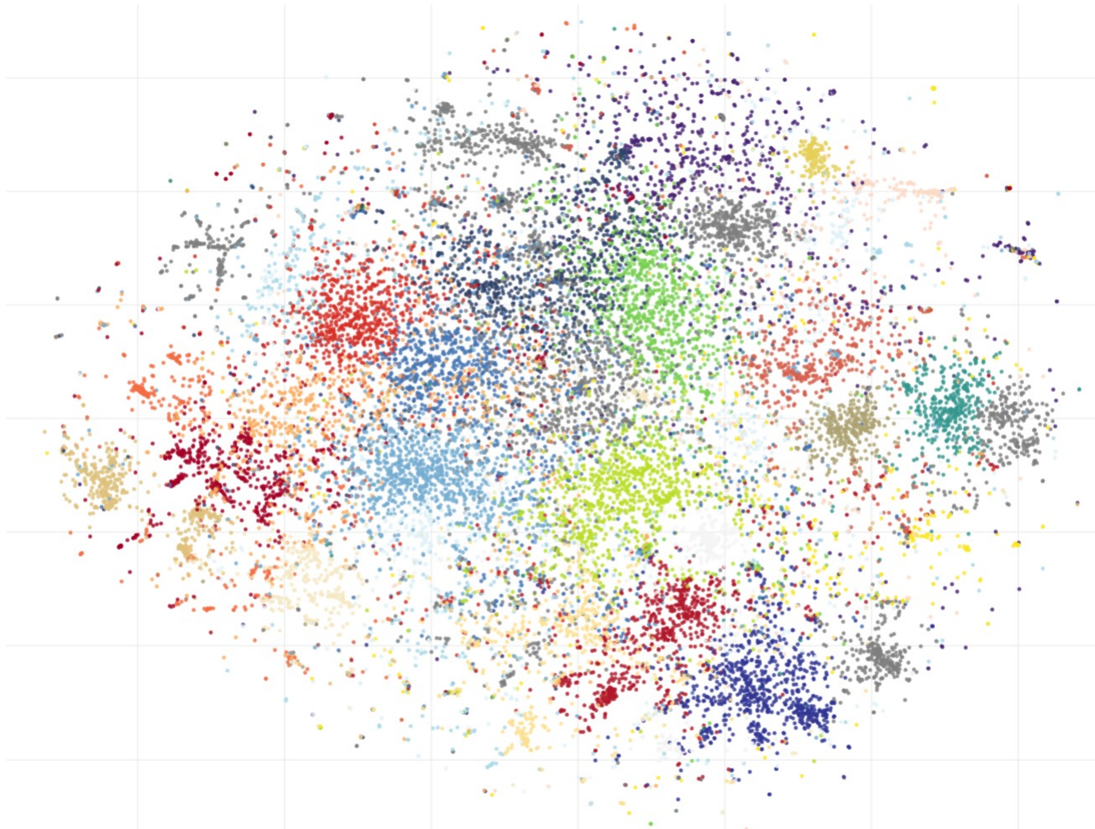


Figure source

Unsupervised learning

- Traditional: Dimensionality reduction, manifold learning
  - Discover a lower-dimensional *latent space* in which the data lives

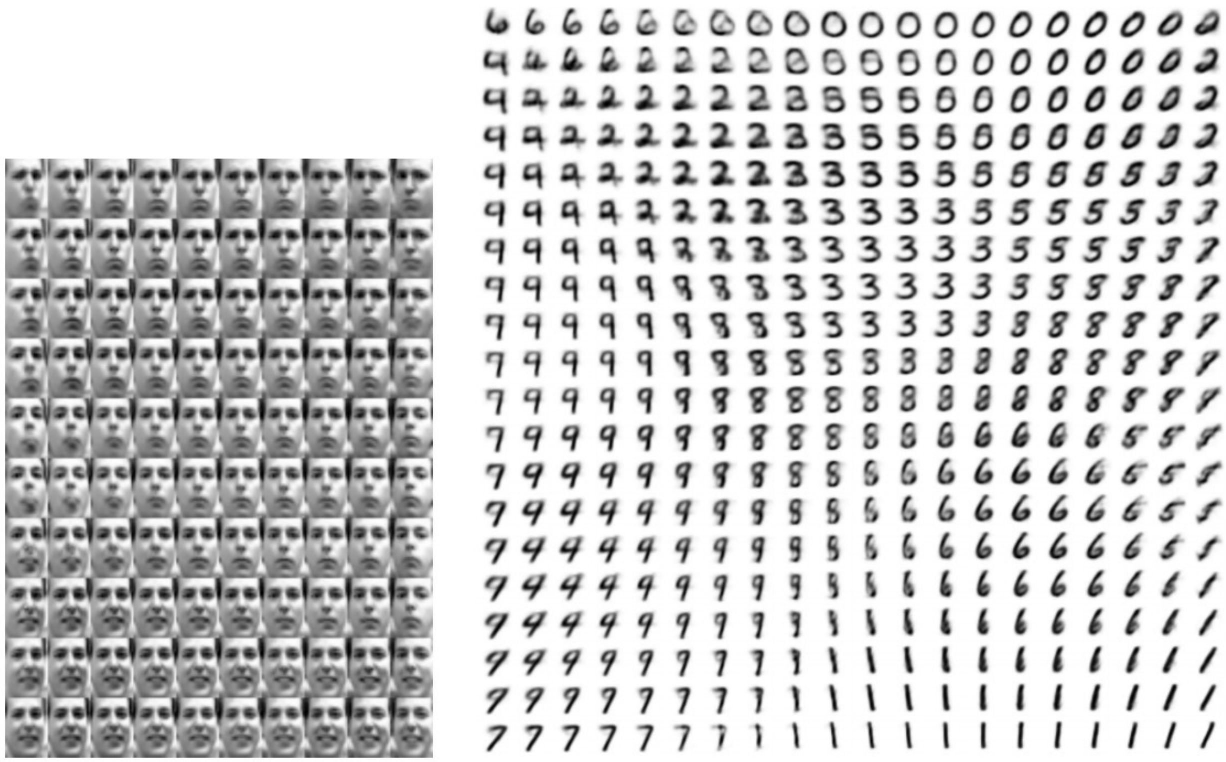
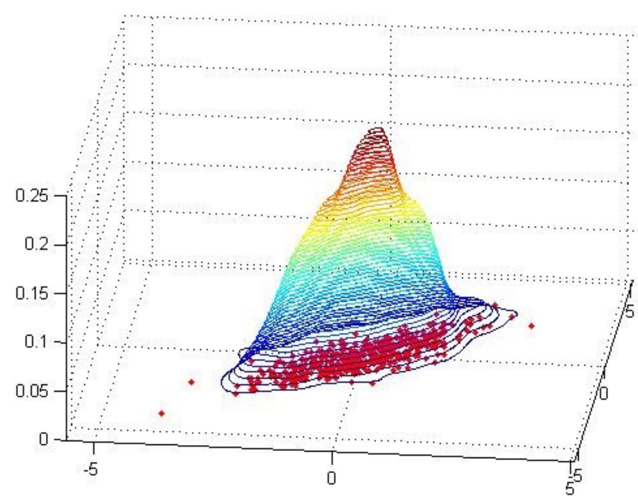


Figure source

Unsupervised learning

- Learning the data distribution
  - **Density estimation**: Find a function that approximates the probability density of the data (i.e., value of the function is high for “typical” points and low for “atypical” points)
  - An extremely hard problem for high-dimensional data…



Unsupervised learning

- Learning the data distribution
  - **Learning to sample:** Produce samples from a data distribution that mimics the training set

Generative adversarial networks (GANs)

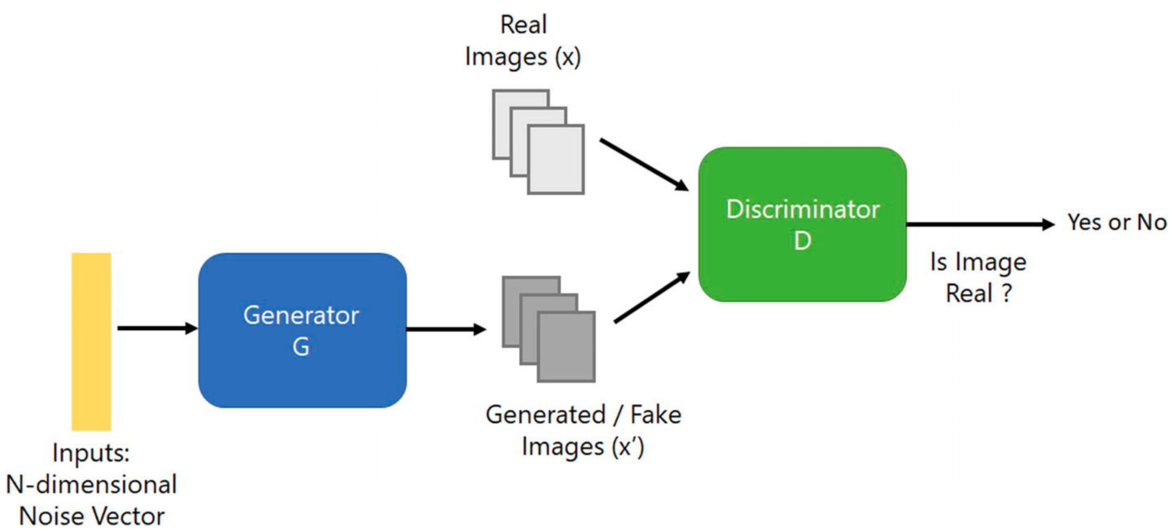
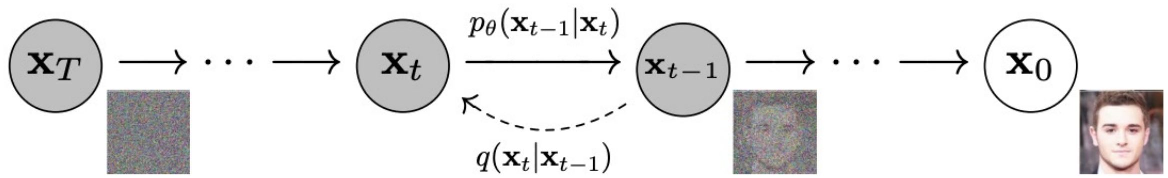


Image source

Unsupervised learning

- Learning the data distribution
  - **Learning to sample:** Produce samples from a data distribution that mimics the training set

Denoising diffusion probabilistic models (DDPMs)



Unsupervised learning

- Learning the data distribution
  - **Learning to sample:** Produce samples from a data distribution that mimics the training set

Denoising diffusion probabilistic models (DDPMs)

vibrant portrait painting of Salvador Dalí with a robotic half face

a shiba inu wearing a beret and black turtleneck

a close up of a handpalm with leaves growing from it

an espresso machine that makes coffee from human souls, artstation

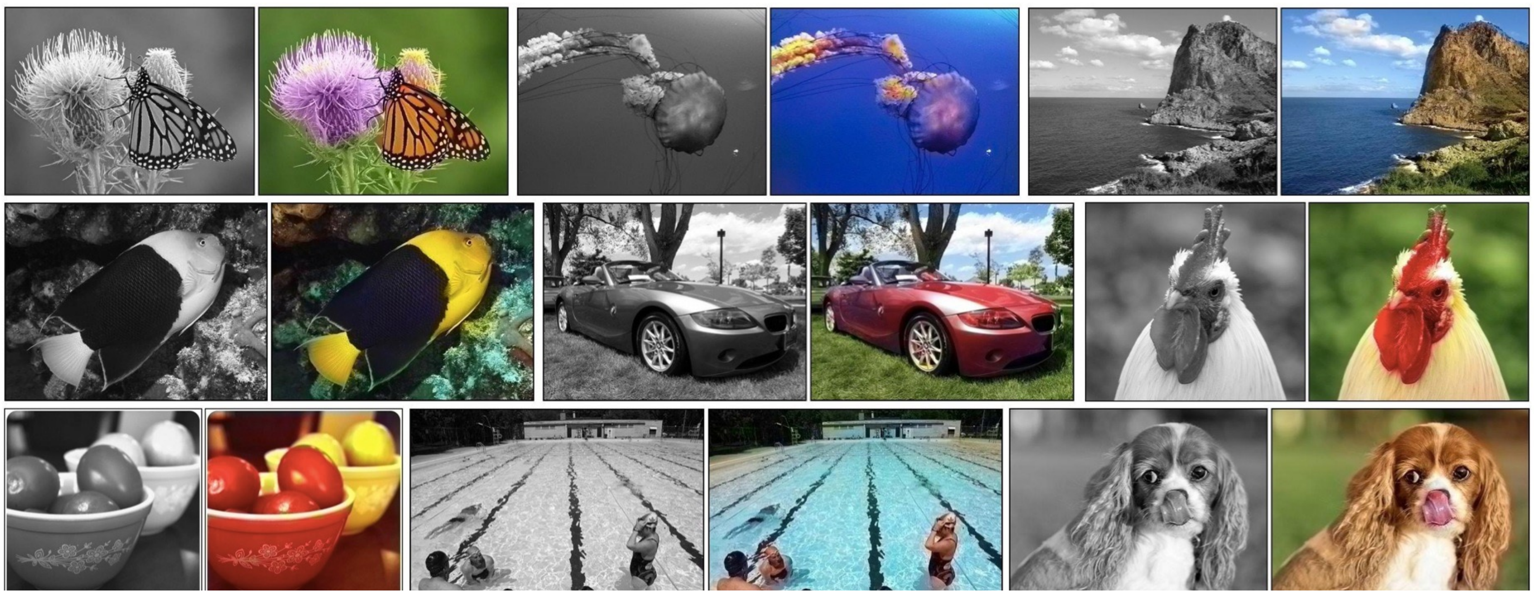
panda mad scientist mixing sparkling chemicals, artstation

a corgi's head depicted as an explosion of a nebula

Source: DALL-E 2

Self-supervised or predictive learning

- Use part of the data to predict other parts of the data
  - Example: Image colorization



R. Zhang et al. Colorful Image Colorization. ECCV 2016

Self-supervised or predictive learning

- Use part of the data to predict other parts of the data
  - Example: Next/masked word prediction

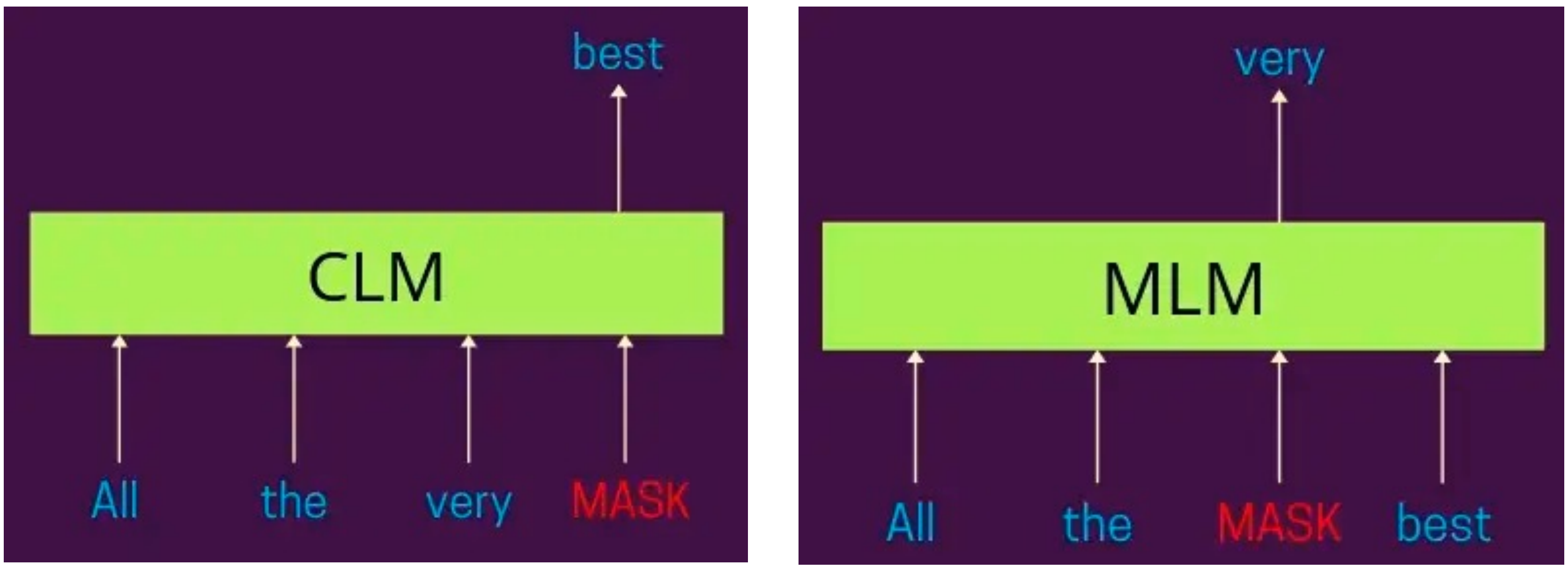
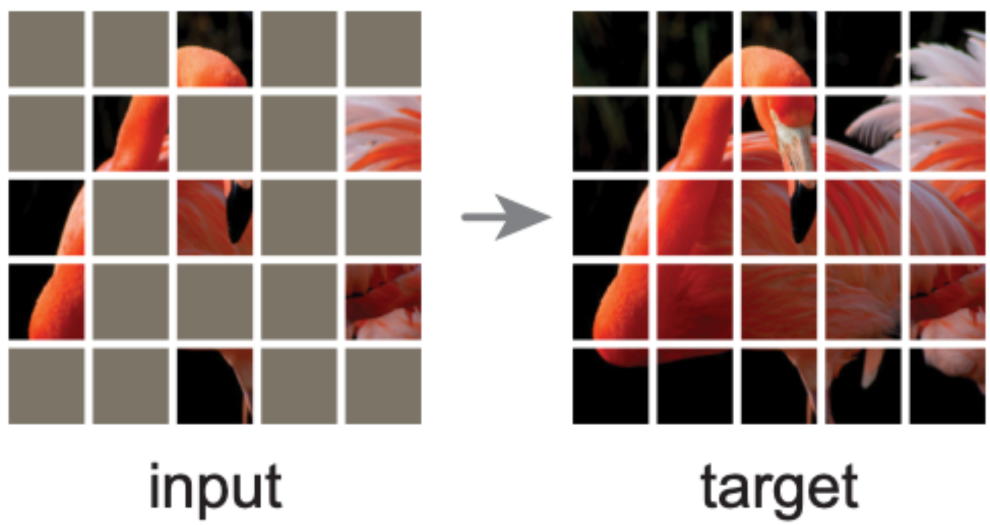


Figure source

Self-supervised or predictive learning

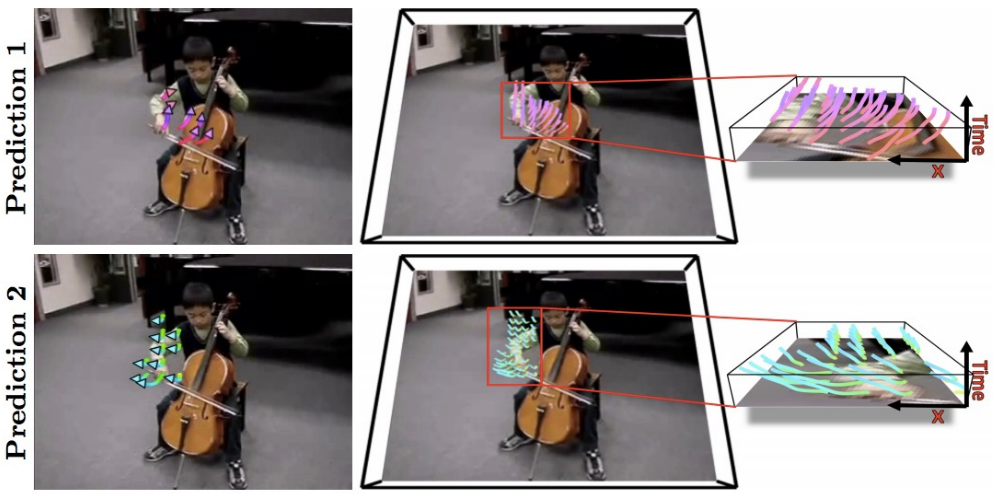
- Use part of the data to predict other parts of the data
  - Example: Masked patch prediction

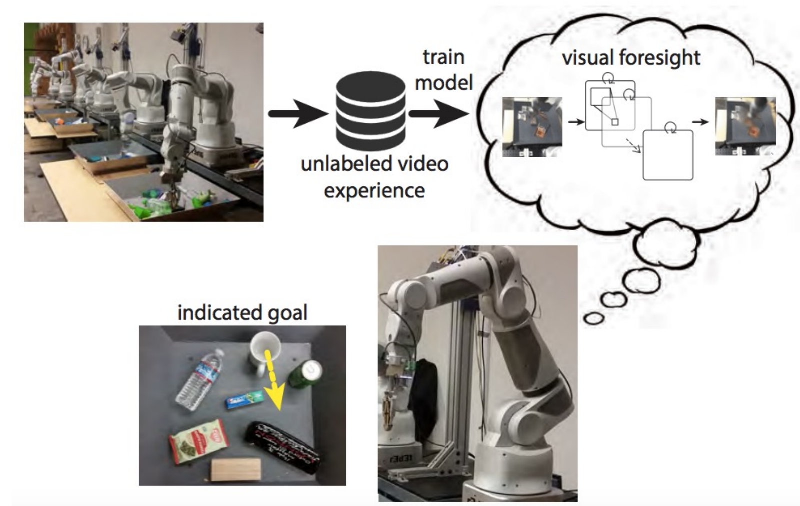


K. He et al. [Masked autoencoders are scalable vision learners](). CVPR 2022

Self-supervised or predictive learning

- Use part of the data to predict other parts of the data
  - Example: Future prediction



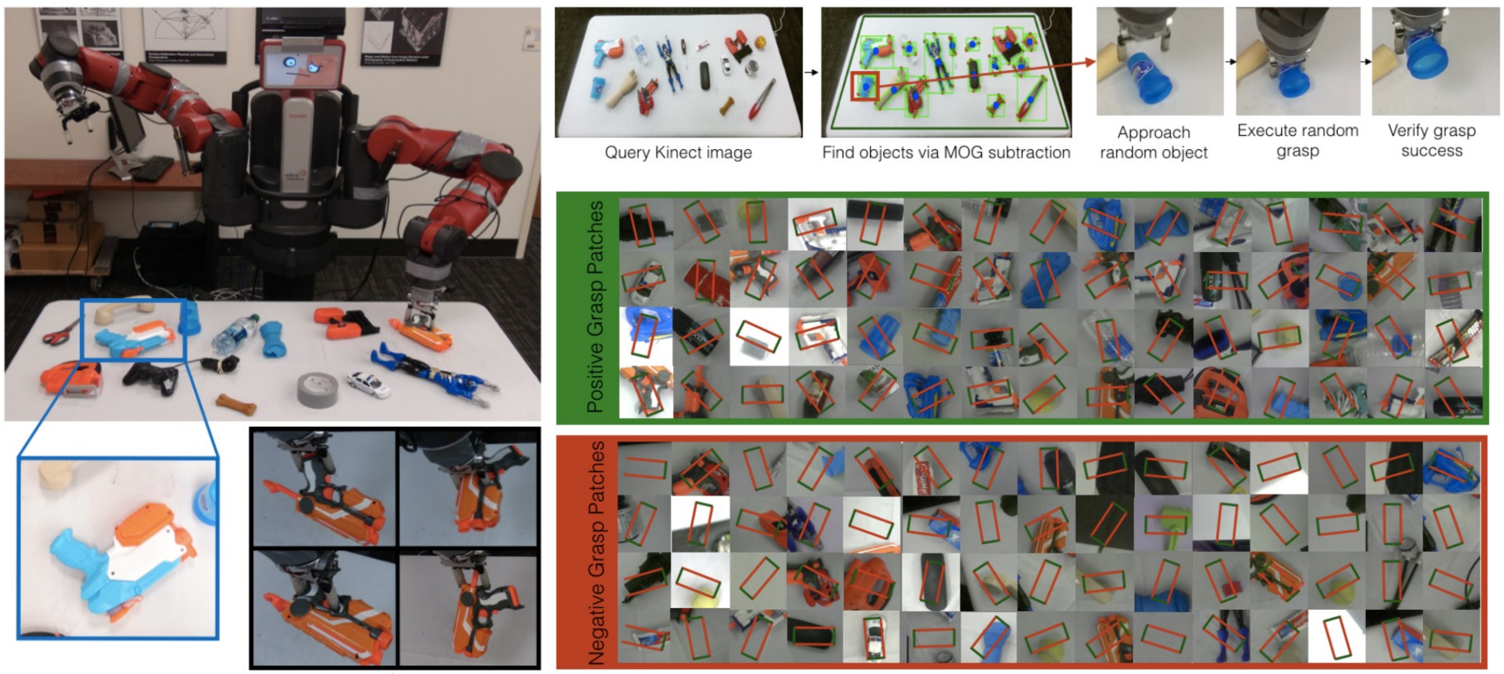


J. Walker et al. An Uncertain Future: Forecasting from Static Images Using Variational Autoencoders. ECCV 2016

C. Finn and S. Levine. Deep Visual Foresight for Planning Robot Motion. ICRA 2017. YouTube video

Self-supervised or predictive learning

- Use part of the data to predict other parts of the data
  - Example: Grasp prediction



L. Pinto and A. Gupta. [Supersizing self-supervision: Learning to grasp from 50K tries and 700 robot hours.]() ICRA 2016

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  - Reinforcement learning

Training regime

Offline learning

Training dataset

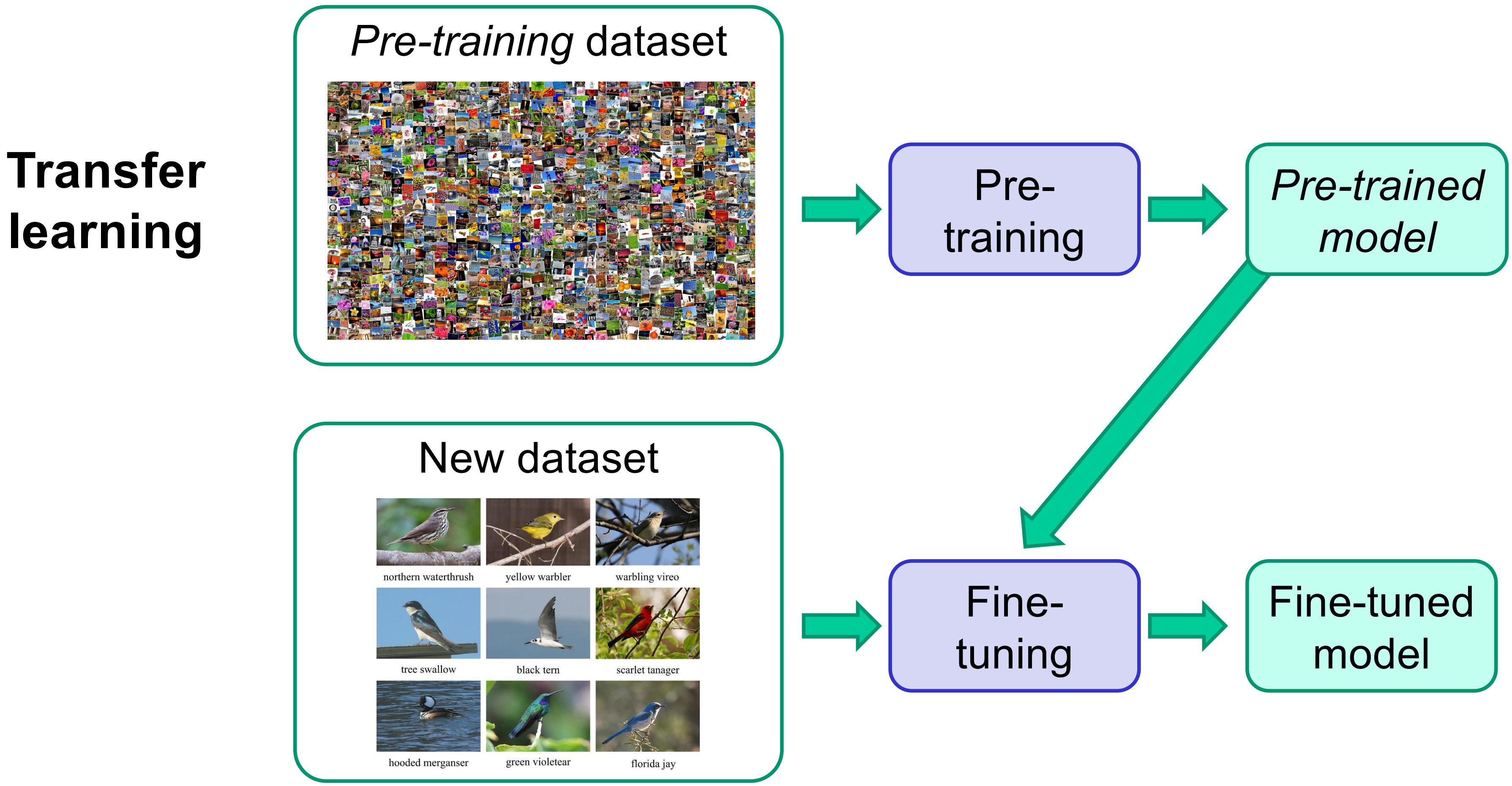
"apple"
"pear"
"tomato"
"cow"
"dog"
"horse"

Process the entire training set
(typically in multiple passes)

Training

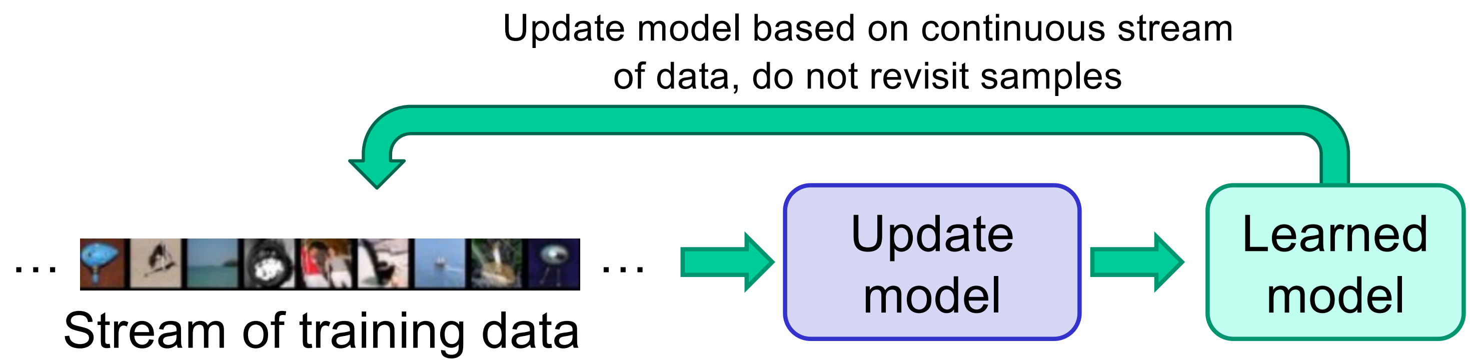
Learned model

Training regime

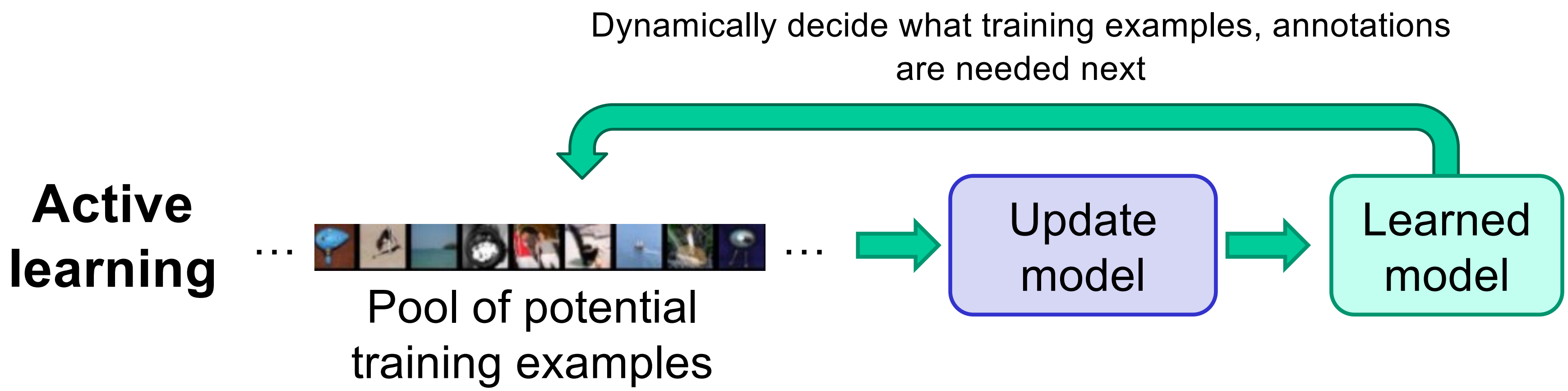


Training regime

Online learning, continual learning



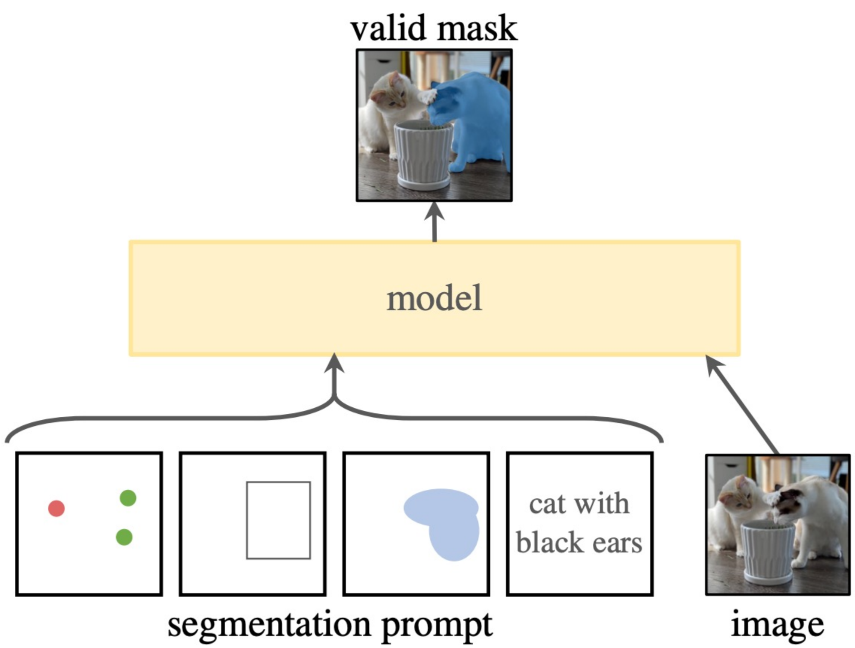
Training regime



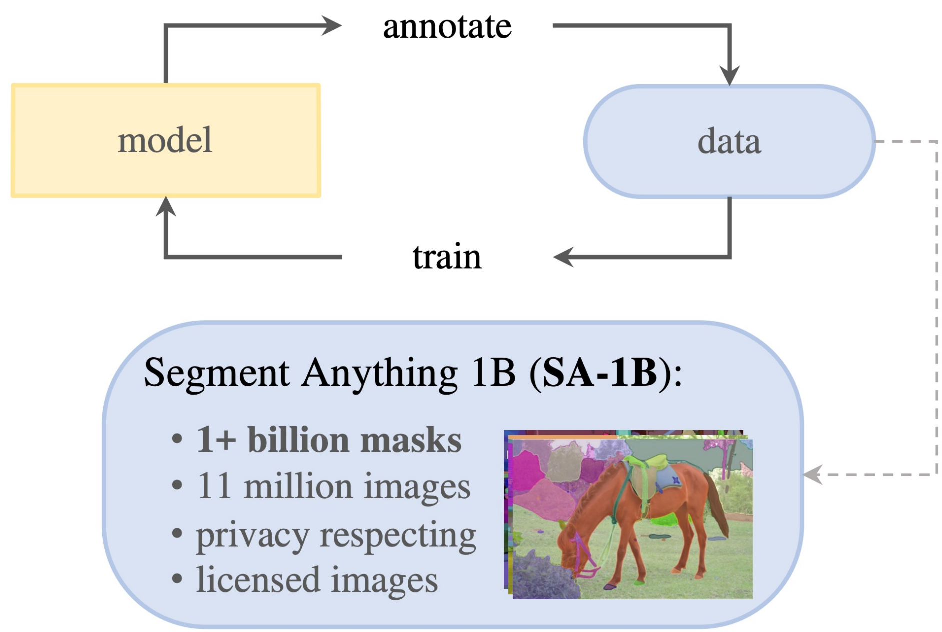
Recent case study: Meta's SAM

- Complex prediction task, hybrid supervision and training regime

Task: Promptable segmentation



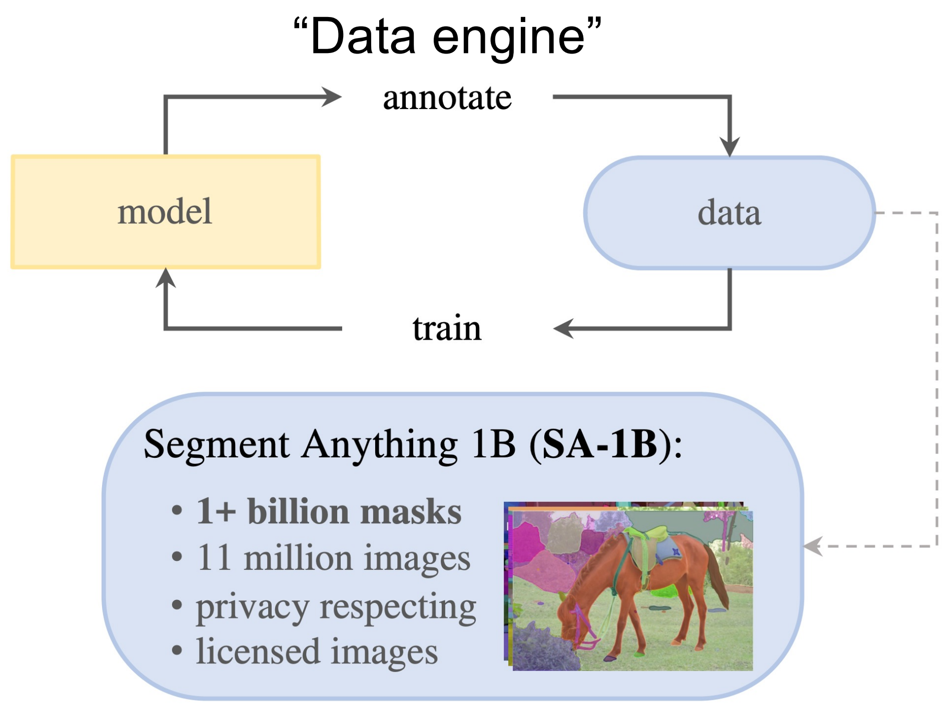
"Data engine"



A. Kirilllov et al. [Segment anything](). ICCV 2023

Recent case study: Meta's SAM

- Complex prediction task, hybrid supervision and training regime
- "Data engine" steps:
  1. **Pre-training** using public datasets
  2. **Assisted manual stage**: interactive segmentation with SAM assisting annotators
  3. **Semi-automatic stage**: SAM generates confident masks, annotators add masks to improve diversity
  4. **Fully automatic stage**: SAM generates ~100 masks per image starting with a grid of points



A. Kirilllov et al. [Segment anything](). ICCV 2023

Reinforcement learning

- Learning for an agent that can affect the world through actions

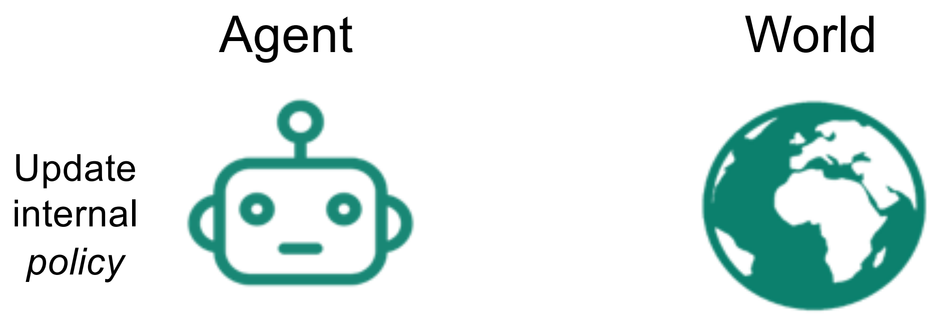
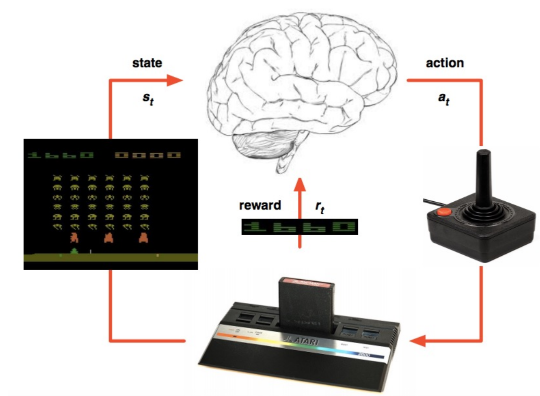


Figure source

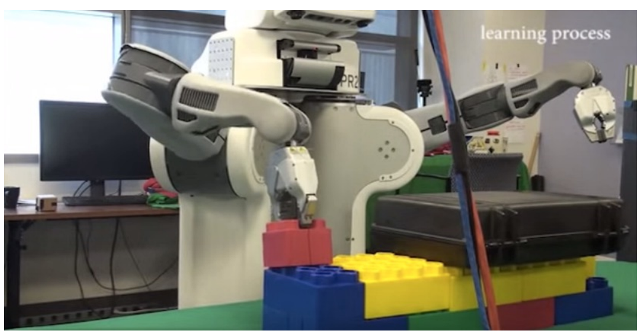
Reinforcement learning: Examples

DeepMind's AlphaGo

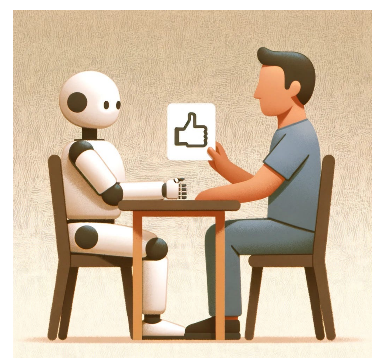
DeepMind's Atari system



Sensorimotor learning



RLHF



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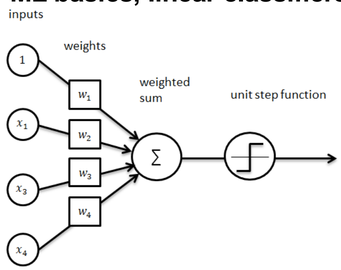
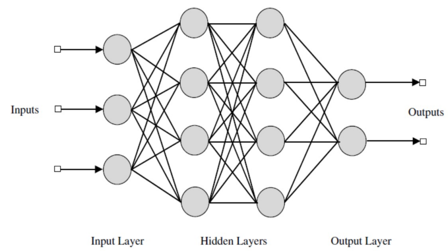
Topics to be covered in class

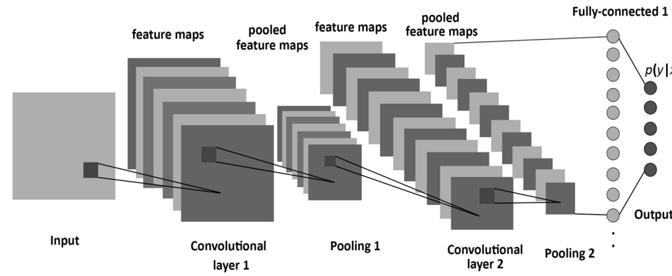
ML basics, linear classifiers



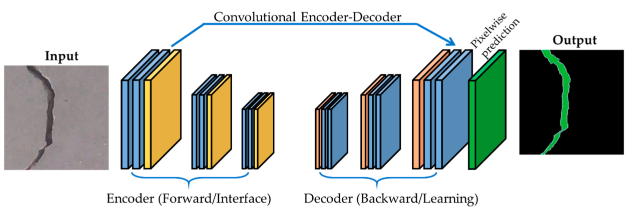
Multilayer neural networks, backpropagation



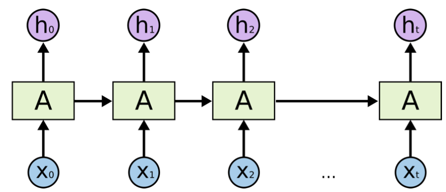
Convolutional networks



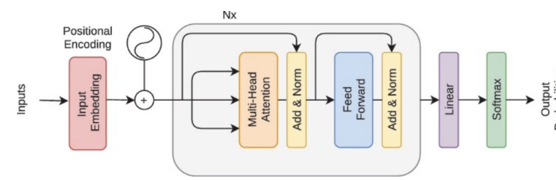
Object detection, dense prediction



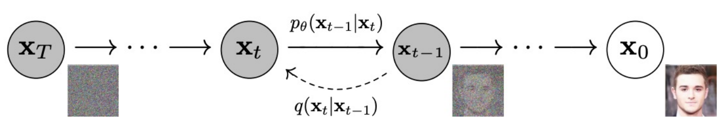
Recurrent networks



Transformers, LLMs, LVMs



Generative models: GANs, image-to-image translation, diffusion models



Deep reinforcement learning